R13

Code No: 111AD

JAWAHARLAL NEHRU TECHNOLOGICAL UNIVERSITY HYDERABAD
B.Tech I Year Examinations, December – 2019/January - 2020
ENGINEERING PHYSICS
(Common to CE, EEE, ME, ECE, CSE, EIE, IT, MMT, AE, AME, MIE, PTM, MSNT, AGE)

Time: 3 hours **Max. Marks: 75**

Note: This question paper contains two parts A and B.
Part A is compulsory which carries 25 marks. Answer all questions in Part A. Part B consists of 5 Units. Answer any one full question from each unit. Each question carries 10 marks and may have a, b, c as sub questions.

PART- A

(25 Marks)

1.a)	Define Ionic bond.	[2]
b)	Explain Burger's vector.	[3]
c)	Name the experiments which supports concept of the dual nature of the light?	[2]
d)	Draw E-K diagram.	[3]
e)	Define Electric Susceptibility.	[2]
f)	Explain Meissner Effect.	[3]
g)	What is Interference?	[2]
h)	Explain Lasing Action.	[3]
i)	Explain the term Nano Scale.	[2]
j)	Distinguish between Direct and Indirect Band gap semiconductors.	[3]

PART-B

(50 Marks)

2.a)	Write a note on Hydrogen Bond.	
b)	Calculate packing factor of BCC.	
c)	Derive an expression of Bragg's Law.	[3+3+4]
	OR	
3.a)	Define Vander-Waal's Bond.	
b)	Estimate Cohesive Energy of diatomic molecule.	
c)	With a diagram explain Powder method.	[2+4+4]
4.a)	Drive an expression of de-Broglie's hypothesis.	
b)	Discuss Density of States.	
c)	What are Matter Waves?	[4+4+2]
	OR	
5.a)	Derive Schrödinger's Time independent wave equation.	
b)	Discuss Kronig-Penny Model.	
c)	Derive an expression of concept of effective mass of an electron.	[3+4+3]
6.a)	Derive an equation for Ionic polarizability.	
b)	What is Hysteresis curve? Explain it using domain theory of ferro magnetism.	
c)	Distinguish between Anti-Ferro and Ferri magnetic materials.	[4+3+3]
	OR	
7.a)	Derive an expression of Clausius - Mossotti equation.	
b)	Distinguish between Soft and Hard Magnetic Materials.	
c)	What is Ferro- electricity? Explain in details.	[4+3+3]

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8.a) In detail discuss Newton rings experiment.
b) With neat diagram explain working principle and construction of He-Ne Laser. [5+5]

OR

9.a) Explain construction and working of Nicol's Prism.
b) Write a note on Semiconductor diode laser. [5+5]

10 .a) Explain working principle and construction of LED.
b) What are the factors affecting the architectural acoustics.
c) Explain the term surface to volume ratio. [4+3+3]

OR

11.a) Define Reverberation of Time.
b) Explain working principle of Photo Diode.
c) Write a note on Quantum Confinement. [3+4+3]

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